Magic, error correction and today's quantum computers

A primer for a QFT / QI-in-QFT physicist, with the links to OE2 — CONICET interview notes, V. Benedetti

How to read this. Sections 1–3 build the quantum-computing (QC) and error-correction (QEC) vocabulary from scratch, using analogies to things you already know: Rényi entropies, replicas, topological order, 1-form symmetries and defects. Section 4 says what the five experiments in the folder actually did. Section 5 is the bridge to OE2. Background definitions are standard textbook material. Statements about the state of the field and the experiments come from the folder papers (sources at the end). Analogies and the few results not taken from the papers are marked as such.

1. The minimal QC vocabulary

Qubits, Paulis, circuits. A quantum computer is a lattice of N two-level systems (qubits). The Hilbert space is $(\mathbb{C}^2)^{\otimes N}$, as in a spin chain, except there is no Hamiltonian to diagonalize. Instead you apply a sequence of few-qubit unitaries ("gates") and then measure in the Z basis. The **Pauli strings** $P = \sigma_1 \otimes \ldots \otimes \sigma_N$, with $\sigma \in \{1, X, Y, Z\}$, are a basis of 4^N operators, the lattice analogue of a complete set of local operators.

Clifford gates and stabilizer states. The *Clifford group* consists of the unitaries that map Pauli strings to Pauli strings under conjugation. It is generated by the Hadamard H (exchanges $X \leftrightarrow Z$), the phase gate S = diag(1, i) and the two-qubit CNOT. A *stabilizer state* is the unique joint +1 eigenstate of N independent, commuting Pauli strings, its "stabilizers". Equivalently, it is what Clifford gates produce from $|00\ldots0\rangle$. Examples: product states, Bell pairs (stabilized by XX and ZZ), GHZ states, and the ground state of the toric code. Stabilizer states can be **highly entangled**, but their entanglement spectrum is always **flat**: all Rényi entropies of a region are equal (the fact used in OE2).

Gottesman–Knill. In the Heisenberg picture, a Clifford circuit maps each stabilizer to another Pauli string. Tracking N Pauli strings costs only poly(N) bits, so stabilizer states + Clifford gates + Pauli measurements are **efficiently simulable classically**, however entangled the state. *Analogy:* this is like free (Gaussian) theories, where Wick's theorem makes everything computable from two-point data. In the same spirit, "magic" plays the role of non-Gaussianity. This is a heuristic, not an exact dictionary.

The missing ingredient: the T gate. Adding a single non-Clifford gate, typically $T = \mathrm{diag}(1, e^{i\pi/4})$ (a "square root of S"), makes the gate set **universal**: any unitary can be approximated, with an overhead logarithmic in the precision (Solovay–Kitaev). The Harvard experiment demonstrated this scaling directly [H]. The classical simulation cost then grows exponentially in the number of T gates, so the non-Clifford content is *the* resource behind quantum speed-ups.

> **Magic, in one line.** Magic is the distance of a state from the set of stabilizer states, i.e. how much non-Clifford resource it carries. One convenient measure, the **stabilizer Rényi entropy**, is $M_n(\psi) = (1-n)^{-1} \log [\, 2^{-N} \Sigma_P \langle\psi|P|\psi\rangle^{2n} \,]$. It vanishes exactly on stabilizer states (2^N Paulis with $\langle P\rangle = \pm 1$) and is a sum over Pauli insertions, built like a replica partition function. OE2 proposes computing it in CFT with topological defects.

Gate teleportation (how magic is "spent"). Hardware does not apply T directly (Section 2 explains why). Instead one prepares the **magic state** $|T\rangle = T|+\rangle = (|0\rangle + e^{i\pi/4}|1\rangle)/\sqrt{2}$ offline and consumes it. Apply a CNOT from the data qubit $|\psi\rangle$ onto the $|T\rangle$ ancilla and measure the ancilla in Z. Outcome 0 leaves $T|\psi\rangle$. Outcome 1 leaves $T^\dagger|\psi\rangle$, which a Clifford correction S turns into $T|\psi\rangle$. Everything except the input $|T\rangle$ is Clifford. So **magic states are a consumable fuel**, made in "magic-state factories", and one is burnt per T gate [G, Go, D]. Kubica and collaborators [G] show that general protocols of this type are equivalent to encoding the input into a stabilizer code followed by a logical non-Clifford gate. They also show that only states Clifford-equivalent to *diagonal* states can be consumed this way, so not every state with magic is usable fuel.

2. Quantum error correction for a QFT physicist

Why we need it. Physical gates fail with probability $p \sim 10^{-3}$ (0.1–0.5% for two-qubit gates on the best devices [P]). Useful algorithms need $\sim 10^9$–10^{12} operations, and Eisert–Preskill estimate about 10^{12} for broad usefulness [P]. Without protection, the state decoheres to the maximally mixed state long before that. "Error mitigation" (classical post-processing of noisy runs) has a cost exponential in circuit size [P]. Hence the current consensus that useful QC means **fault tolerance**, which is also Lucas's point that NISQ is a dead end.

Encoding. One *logical* qubit is stored in a 2-dimensional subspace (the **code space**) of n physical qubits. This subspace is the joint +1 eigenspace of a set of commuting Pauli **stabilizers**, so the encoded states are themselves stabilizer states. One repeatedly measures the stabilizers. An error flips some of them, and the pattern of flips (the **syndrome**) is fed to a classical **decoder** that infers and undoes the most likely error. The notation [[n, k, d]] means n physical qubits, k logical qubits and **distance** d, the minimal number of single-qubit errors that can corrupt the logical information undetected. A distance-d code corrects $\lfloor (d-1)/2 \rfloor$ errors.

The QFT picture: the toric/surface code is a Z_2 gauge theory. Put qubits on the links of a square lattice. Stabilizers are the star operators $A_v = \Pi X$ (Gauss law) and the plaquettes $B_p = \Pi Z$ (zero flux). The code space is the ground space of $H = -\Sigma A_v - \Sigma B_p$: the **deconfined phase of Z_2 gauge theory**, with topological order. On a torus it is 4-fold degenerate (2 logical qubits). The **logical operators are Wilson and 't Hooft loops** around non-contractible cycles, the charged objects of the 1-form symmetry. A local error creates a pair of anyons (e or m), and decoding means pairing the anyons back up. The distance d is the length of the shortest non-contractible loop, which is why local noise cannot see the logical information: it is stored *topologically*, like the long-range entanglement and topological entanglement entropy of AE2. The planar version (surface code) uses $\sim d^2$ qubits per logical qubit. *Color codes* are a cousin defined on 3-colorable lattices. The smallest 2D one is the 7-qubit Steane code [[7,1,3]]; the smallest 3D one is the 15-qubit Reed–Muller code [[15,1,3]].

Threshold theorem. If p is below a threshold p_{th}, increasing d suppresses logical errors exponentially. For the surface code, $P_L \approx 0.1\,(p/p_{th})^{(d+1)/2}$ with $p_{th} \approx 10^{-2}$ [P]. *Analogy (standard, not from the folder):* the threshold is a genuine phase transition of a disordered stat-mech model, and below it the topological phase survives the noise. The price is overhead. At $p = 10^{-3}$, reaching $P_L \sim 10^{-11}$ needs $d \approx 19$, about 10^3 physical qubits per logical qubit, or $\sim 10^6$ physical qubits for a useful machine [P]. Today's devices have hundreds.

Transversal gates and why T is hard. The cheapest fault-tolerant logical gates are *transversal*: apply the same one-qubit gate to every physical qubit (or pair qubit i of one block with qubit i of another). Such gates cannot spread one error into many. They are the on-site, symmetry-like operations of the code, acting logically like a topological defect swept across the whole system. The **Eastin–Knill theorem** forbids any code from having a universal set of transversal gates [G, H]. In practice 2D codes (surface, Steane) give all Clifford gates cheaply but no T. The 3D Reed–Muller code has a transversal T (apply T or $T^\dagger$ on alternating qubits) but lacks other gates [D, H]. *Standard result, not from the folder (Bravyi–König):* for topological codes, the dimension of space limits which logical gates locality-preserving circuits can implement. This is a strong hint that the T bottleneck is a property of the topological phase itself.

Figures of merit used in the experiments. *Fidelity* $F = \langle T|\rho|T\rangle$ of the produced state, and *infidelity* $1-F$. *Break-even*: the encoded state beats the same state made on bare physical qubits. *Pseudo-threshold*: the logical error is below the physical error of the components used. *Post-selection / acceptance rate*: runs in which any check fired are discarded. This is acceptable for magic states, which can be prepared offline and retried, but not for the computation itself [D, Go, P].

3. Making good magic states: the four strategies

- **Injection + distillation.** Encode a noisy physical $|T\rangle$ into a code ("injection"). This is not fault-tolerant, so the result has error $\sim p$. Then *distill*: combine several noisy copies with Clifford operations and post-select on a syndrome, getting fewer but better copies. $15 \to 1$ distillation takes error ε to $\sim 35\varepsilon^3$ [D]. This is robust but very expensive. Used by QuEra [Q].
- **Error-suppressed encoding (flags).** Prepare the magic state through a measurement whose failures are caught by extra "flag" qubits, so that error is $\sim p^2$ instead of $\sim p$. It is cheaper and produces better inputs for distillation. Used by IBM [I].
- **Code switching.** Prepare $|T\rangle$ in the 3D Reed–Muller code, where T is transversal, then teleport it into the 2D Steane code, where all Cliffords are transversal. Teleporting between codes evades Eastin–Knill. Harvard notes that code switching and logical teleportation are the same thing [H]. Used by Quantinuum and Harvard [D, H].
- **Cultivation.** Inject into a small color code, then repeatedly measure the logical Hadamard H_L, of which $|T\rangle$ is an eigenstate. Each fault-tolerant check that passes "grows" the fidelity, and failures are discarded. The state is finally transferred ("grafted") into a larger surface code. Error scales as $\sim p^3$. Used by Google [Go].

4. The experiments in the folder, in one line each

Three hardware platforms dominate [P]. **Superconducting circuits** (Google, IBM): each qubit is an artificial atom, a "transmon" built from a Josephson junction, on a chip at mK. Gates are microwave pulses taking tens of ns, but qubits couple only to nearest neighbours on a 2D grid. **Trapped ions** (Quantinuum): the qubit is two internal levels of a charged atom held in an electromagnetic trap. Gates are laser pulses entangling ions through their shared vibrational modes; ions can be shuttled around, giving effectively all-to-all connectivity and very low errors, but gates take tens of µs. **Neutral atoms** (Harvard, QuEra): Rb atoms held in optical tweezers. They are entangled by briefly exciting them to Rydberg states with strong dipole interactions, and whole code blocks are moved around by moving the tweezers. This platform has many qubits and flexible connectivity, but it is slow.

Experiment	Platform / size	What was done (one sentence)	Take-home
IBM 2023 [I]	Superconducting; 7 qubits of a 27-qubit chip	A 2-qubit magic state ($\|CZ\rangle$ $\|00\rangle+\|01\rangle+\|10\rangle$) prepared inside a 4-qubit error-detecting code by a flagged measurement, using mid-circuit measurement and feed-forward.	First encoded magic state **better than the best physical one** on the chip (break-even), at $\sim$1% error.
QuEra 2024 [Q]	Neutral atoms; 35 (d=3) to 85 (d=5) atoms = 5 logical qubits	5 noisy logical magic states in color codes are distilled into 1 better one with transversal Clifford gates.	First distillation **at the logical level**; error reduced $\sim$6–8×.

Experiment	Platform / size	What was done (one sentence)	Take-home
Harvard 2025 [H]	Neutral atoms; up to 448	A complete fault-tolerant toolkit: below-threshold surface code, transversal T via 15-qubit 3D codes plus teleportation, and deep circuits with atom reuse.	All ingredients of universal FT computation in one machine; **measured the magic** (Bell magic) of the encoded state.
Quantinuum 2025 [D] (Lucas)	Trapped ions; 28 qubits	Code switching from the 15-qubit 3D code to the 7-qubit Steane code, certified with a two-copy measurement.	Error $\sim 5\times10^{-4}$, **below the physical gate error**; completes a universal logical gate set at or below physical error.
Google 2025 [Go]	Superconducting (Willow); ∼54 qubits	Cultivation in a small (distance-3) color code, then grafting into a distance-5 surface code.	Error $\sim 10^{-4}$, 40× better than injection; 8% of attempts kept.

Where this leaves us. Since 2023 every major platform has made logical magic states better than the physical operations that produce them, reaching errors of $\sim 10^{-4}$. Targets are $\sim 10^{-6}$–10^{-8} for quantum simulation and $\sim 10^{-10}$ for factoring [Go], so one or two more orders of magnitude, or one extra round of distillation, are needed. Remaining problems: heavy post-selection that does not scale, certifying such small errors (naive tomography needs $\sim 10^{8}$ shots for 10^{-4}, so the papers invent two-copy and "kickback" estimators [D, Go]), decoding speed, and growing from $\sim 10^{2}$ to $\sim 10^{6}$ qubits [P]. Eisert–Preskill expect the first useful fault-tolerant tasks ("megaquop", 10^{6} operations) to be *scientific*, notably simulating strongly coupled lattice systems. About 18% of US DOE supercomputer time goes to lattice QCD [P].

5. Why this matters for OE2

- **Magic is the expensive resource in real hardware.** Section 1 explained why magic, not entanglement, measures classical non-simulability. Section 4 shows the whole engineering effort exists to make it. Magic measures cannot increase under Clifford operations. The magic of a target state, e.g. the vacuum of a lattice model near criticality, therefore sets a lower bound on how many magic states a fault-tolerant simulation must consume. Computing magic of QFT/CFT states is thus a theoretical estimate of a concrete cost.
- **Magic vs. entanglement, seen experimentally.** OE2: stabilizer states have flat entanglement spectra, so a non-flat spectrum lower-bounds magic. Harvard [H] finds the converse face of this: *logical magic requires physical entanglement*. $|T_L\rangle$ is an eigenstate of $(X_1X_2\ldots + Y_1Y_2\ldots)/\sqrt{2}$, a superposition spread across the whole code block. Since Clifford decoding conserves magic, 15 physical T gates yield only one unit of logical magic, which is possible only through entanglement.
- **Replica-type magic measures are measured in the lab.** Rényi entropies need n copies, which is the replica trick in QFT and literally n copies of the state in experiment. Harvard measured the two-copy "Bell magic" (Haug–Kim), a cousin of the stabilizer Rényi entropies, on the 15-qubit code, finding exactly one unit of magic after a transversal T [H]. Quantinuum certified its state with a two-copy (purity, Rényi-2-like) measurement [D]. The objects OE2 computes in CFT (partition functions on replica manifolds with Pauli or defect insertions) have direct experimental counterparts.
- **Topological codes vs. continuum QFT.** The codes are ground states of lattice topological phases. They are exactly the setting where stabilizer states exist, as OE2 notes for spin chains and topological theories. By contrast, in continuum QFT with local degrees of freedom (type III_1 local algebras) the work in progress argues every state carries magic. Logical gates act through the code's topological structure: cheap Cliffords in 2D, T only in 3D, and teleportation between them. *My interpretive remark:* this is the natural place where the defect / generalized-symmetry language of OE1–OE3 can speak to QC.
- **Which magic is useful?** Not every non-stabilizer state can be spent as fuel [G]. For example, the state that QuEra distills is not usable in the standard teleportation scheme. This supports OE2's focus on *which* magic measure is physically meaningful.

6. Pocket dictionary

QC / QEC term	What it is	Closest QFT / QI analogue
Pauli string	Product of X, Y, Z, 1 on sites	Complete basis of local operators
Clifford gate	Maps Paulis to Paulis	Free/Gaussian (Bogoliubov-like) transformation (heuristic)
Stabilizer state	Joint eigenstate of N commuting Paulis	Classically tractable state with flat entanglement spectrum
Magic	Non-stabilizerness	Non-Gaussianity / “interaction content” (heuristic)
Code space	Joint +1 eigenspace of stabilizers	Degenerate ground space of a topological phase
Logical operator	Acts within code space	Wilson / 't Hooft loop (1-form symmetry operator)
Error, syndrome	Local fault, its detected signature	Anyon pair created, positions of anyons
Distance d	Min. weight of a logical operator	Shortest non-contractible loop (system size)
Threshold	Max. tolerable physical error	Phase transition out of the topological phase
Transversal gate	Same 1-qubit gate on every site	On-site symmetry / defect swept through space
Gate teleportation	Spend $\|T\rangle$ to apply T	Resource consumption (resource theory)
Two-copy measurement	Measure $\rho\otimes\rho$ jointly	Replica trick with n = 2

Sources (folder papers): [P] Eisert & Preskill, arXiv:2510.19928. [Go] Rosenfeld et al. (Google), arXiv:2512.13908. [Q] Sales Rodriguez et al. (QuEra), arXiv:2412.15165. [H] Bluvstein et al. (Harvard), arXiv:2506.20661. [I] Gupta et al. (IBM), arXiv:2305.13581. [D] Daguerre et al. (Quantinuum), arXiv:2506.14169. [G] Zheng, Zang & Kubica, arXiv:2607.08508. OE2 statements: CIC 2025 research plan. Textbook background (Pauli/Clifford/stabilizer formalism, Gottesman–Knill, toric code, Solovay–Kitaev) and the explicitly flagged analogies and results (Gaussian analogy, stat-mech threshold, Bravyi–König) are not taken from the folder.